MIL-S-43336C
3 March 1978
SUPERSEDING
MIL-S-43336B
27 August 1970

MILITARY SPECIFICATION

SEALING IRONS AND SEALING PLATES, ELECTRIC,

MANUALLY OPERATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hand operated electric sealing irons and plates (see 6.1).

1.2 Classification. Sealing irons and plates shall be of the following types as specified (see 6.2):

Type I - Sealing iron, 125 to 150 watts capacity w/rest.
Type II - Sealing plate, 900 to 1000 watts capacity.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-26 - Bag, Plastic (General Purpose)
PPP-B-566 - Boxes, Folding, Paperboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3540

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PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-676 - Boxes, Setup

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNDERWRITERS LABORATORIES, INC. (UL)

UL Standard 499

(Application for copies should be addressed to the Underwriters Laboratories Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingsten Road, Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, LI, NY 11746; 1655 Scott Blvd., Santa Clara, CA 95050; or 2602 Tampa East Blvd., Tampa, FL 33619.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

Standard No. 70 - The National Electrical Code

(Application for copies of the National Electrical Code should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Standard product. Sealing irons and sealing plates shall be the manufacturer's regular commercial products except for any changes necessary to comply with specification requirements.

3.2 Code and standards.

3.2.1 Compliance. The electrical materials and components of the sealing irons and plates shall conform to UL Standard 499. Installation of the electrical components of the sealing irons and sealing plates shall conform to the applicable requirements of NFPA Standard No. 70.

3.2.2 Certification. Prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the sealing irons and sealing plates he proposes to furnish under this specification meet the requirements specified in 3.2.1.

3.2.2.1 UL. Acceptable evidence of meeting the requirements of Underwriters Laboratories, Inc. shall be the UL Label or Listing Mark, or a certified test report from a recognized independent testing laboratory acceptable to the Government, indicating that the electrical materials and components offered have been tested and conform to UL Standard 499.

3.2.2.2 NFPA. Acceptable evidence of meeting the requirements of NFPA Standard No. 70 shall be the manufacturer's certified statement that the sealing iron and sealing plate conform to applicable NFPA requirements.

3.3 Materials. Materials shall conform to the requirements specified hereinafter. Materials not definitely specified shall be of the quality normally used by the manufacturer for his sealing irons and sealing plates provided the completed items comply with all the provisions of this specification (see 6.3).

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3.4 Design and construction. The type I sealing iron and the type II sealing plate shall have heat control by thermostatic means. At an ambient temperature between 68°F and 72°F (20°C and 22°C) the sealing iron and sealing plate shall have (1) a capability of maintaining a uniform temperature on the sealing surface within ± 40°F (22°C) of the preset temperature when tested as specified in 4.3.1, and (2) a heat recovery capability of 50°F (28°C) ± 15°F (8.5°C), within 80 seconds when tested as specified in 4.3.2.

3.4.1 Metal components. The sealing surfaces of the sealing iron and sealing plate shall be either corrosion-resisting steel or brass.

3.4.2 Electric cord. The sealing iron and sealing plate shall be provided with a suitable cord of 6 feet (1.8 m) minimum length and plug. The cord shall be provided with a resilient type adapter to prevent sharp bends in the cord when it is attached to the sealing unit.

3.4.3 Thermostat. The thermostat shall be graduated in temperature intervals of 25°F (14°C) or less over a temperature range from 200°F to 500°F (93°C to 260°C). The temperature of the heating surfaces shall be controlled within 40°F (22°C) for temperatures from 200°F to 300°F (93°C to 149°C) and within 25°F (14°C) for temperatures from 301°F to 500°F (149°C to 260°C) when tested as specified in 4.3.4.

3.4.4 Electrical characteristics. Unless otherwise specified (see 6.2), the sealing iron and sealing plate shall operate from a 120-volt, 60 Hz single phase, power supply.

3.4.5 Dielectric strength and insulation requirements.

3.4.5.1 Dielectric strength. The sealing iron and sealing plate shall be capable of withstanding for 1 minute without breakdown a 60 Hz essentially sinusoidal potential of 1 kv applied between live metal parts and dead metal parts in accordance with Underwriters Laboratory Standard UL-499.

3.4.5.2 Insulation. The sealing iron and sealing plate shall have an insulation resistance of not less than 2 MΩ between live metal parts and interconnected dead metal parts in accordance with the requirements of Underwriters Laboratory Standard UL-499.

3.4.6 Type I. The sealing iron shall have a handle and stem resembling a soldering iron. The heating surface shall be 3-7/8 inches long by 1-3/8 inches wide tapering to 1-1/8 inches, within a tolerance of ± 1/16 inch, (98 mm long by 35 mm wide, tapering to 28 mm, within a tolerance of ± 1.6 mm).

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The stem and handle shall be capable of withstanding a total force of 5 lbs (22.2 N) with no visual evidence of deflection or permanent deformation when tested as specified in 4.3.5. The handle shall not exceed a temperature of 110°F (43°C) after the iron has been heated at a 450°F (232°C) temperature for 1 hour in a 68°F to 72°F (20°C to 22°C) ambient temperature when tested as specified in 4.3.6. The total weight of the iron and cord shall not exceed 1-1/2 lbs (.7 kg). The wattage of the sealing irons shall be between 125 and 150 W when tested as specified in 4.3.7. When specified (see 6.2) each sealing iron shall be provided with a polytetrafluoroethylene shoe cover.

3.4.6.1 Rest. The sealing iron shall be provided with a rest fabricated from corrosion-resisting steel sheet or wire for resting the iron during use. The rest shall provide effective insulation so that the rest with the iron at 450°F (232°C) operating temperature placed over a sheet of white bond paper for 1 hour shall produce no discoloration to the paper when tested as specified in 4.3.8.

3.4.7 Type II. The sealing plate shall be rectangular in configuration and shall consist of a metal top plate with one or more strips of electrical heating elements inclosed and shielded by a metal casing. The casing shall be smooth and recessed a minimum of 1/16 inch (1.6 mm) from all outer edges of the heating surface. The sealing plate shall have four stable legs in accordance with the manufacturer's commercial practice and means for flush-mounting the unit on a flat surface. The wattage of the sealing plate shall be between 900 and 1 kw when tested as specified in 4.3.7. The dimensions of the sealing plate shall be 12 inches long by 9 inches wide within a tolerance of $\pm$ 1/16 inch (304 mm by 229 mm $\pm$ 1.6 mm). When specified (see 6.2) the sealing plate shall be provided with a polytetrafluorethylene cover for this heating surface.

3.5 Finish. The sealing iron and sealing plate shall be finished in accordance with the manufacturer's commercial practice.

3.6 Identification marking. Each sealing iron and sealing plate shall be marked for identification in accordance with the manufacturer's commercial practice. The marking shall be complete and legible.

3.7 Workmanship. Workmanship of the sealing iron and sealing plate shall exhibit in particular the following:

(1) No missing components.
(2) Free of fractured, broken, dented or loose components.
(3) Free of sharp edges, burrs, and slivers.
(4) Components in alignment.
(5) Integrity of electric cord and plug.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contrac the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 End item inspection. A lot shall consist of all sealing irons or all sealing plates offered for inspection at one time. The sample unit shall be one completely fabricated sealing iron or plate.

4.2.2.1 Visual examination. Visual examination shall be made of the end item for defects in table I. The inspection level shall be II and the acceptable quality level (AQL) shall be 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

TABLE I. Visual examination

Examine	Defect	Classification Major	Minor
Finish	Not manufacturers commercial finish	X	
Design and construction (general)	Not manufacturers commercial item	X	

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TABLE I. Visual examination (cont'd)

Examine	Defect	Classification Major	Minor
Workmanship (applicable to unit and component parts)	Unit not free from sharp edges, burrs, and slivers		X
	Any component parts loose		X
	Any component missing, fractured, or dented	X	
	Any component not in alignment	X	
	Wire broken or plug disconnected from cord	X	
	Plug prong is loose, bent, or twisted		X
	Cord insulation cracked, cut, frayed, or abraided	X	
Marking for identification	Missing, incomplete, or not legible	X	

4.2.2.2 Dimensional examination. Examination shall be made of the end item for compliance with dimensions specified. Any nonconformance shall be classified a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

4.2.2.3 End item testing. Sealing irons and sealing plates shall be tested in accordance with 4.3.1 thru 4.3.8 as applicable. The inspection level shall be S-2 with an AQL of 2.5 expressed in terms of defects per hundred units.

4.2.3 Code and standard compliance. Proof of compliance with the requirements of 3.2.1 shall be available to the Government representative.

4.2.4 Examination of packaging inspection. An examination will be made to determine that preservation-packaging, packing and marking required by section 5 are complied with. Defects will be recorded as listed in table II. The sample unit shall be one shipping container fully prepared for delivery except that it need not be closed. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the AQL shall be 4.0 expressed in terms of defects per hundred units.

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TABLE II. Inspection of packaging

Examine	Defect
Marking, exterior and interior	Incorrect; illegible; omitted; of improper size, location, sequence, or method of application
Materials	Component missing, damaged
Workmanship	Bulging or distorted containers
Content	Number per container is more or less than specified

4.2.4.1 Inspection of interior boxes. When interior boxes are required to comply with PPP-B-566, PPP-B-636 or PPP-B-676, examination of defects in the closure shall be in accordance with the appendix of the applicable box specification.

4.2.4.2 Inspection of shipping containers. When shipping containers are required to comply with PPP-B-636, examination for defects in the closure, waterproofing and reinforcing shall be in accordance with the appendix of PPP-B-636.

4.3 Tests.

4.3.1 Uniform temperature test. At an ambient temperature between 68°F and 72°F (20°C and 22°C) and with the thermostat set at 300°F (150°C), the sealing iron or plate, as applicable, shall be heated for a minimum of 30 minutes for temperature stabilization. The temperature of the sealing surface shall be checked at five points as follows: The points shall be located on the longitudinal and minor axis intersecting at the geometric center of the heating surface, with one point at the geometric center. Each axis shall have a point located at a distance from each end, of 1/2 inch (13 mm) for the longitudinal axis and 1/4 inch (6 mm) for the minor axis. The temperature at each of the five points shall be checked against the average temperature obtained to determine compliance with the uniform temperature requirements of 3.4. Any nonconformance shall constitute failure of this test.

4.3.2 Heat recovery test. At an ambient temperature between 68°F and 72°F (20°C and 22°C) and by means of the thermostat, the sealing iron or sealing plate, as applicable shall be heated until temperature stabilization at 450°F (232°C) has been reached. The sealing surface temperature shall be cooled 50°F (28°C) and then allowed to heat until the temperature is again 450°F (232°C). The time required to increase the temperature shall then be checked. Nonconformance with the heat recovery requirements of 3.4 shall constitute failure of this test.

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4.3.3 Dielectric strength and insulation tests. These tests are to be conducted with the operating switch turned "ON" and any nonconformance shall constitute failure of the test.

4.3.3.1 Dielectric strength test. The unit shall be tested for compliance with the requirements of 3.4.5.1 by means of a suitable 500 volt-ampere or larger capacity testing transformer, the output voltage of which is essentially sinusoidal and capable of being varied. The applied potential shall be increased gradually from 0 until the required test value of 1 kv (RMS) is reached and held at that value for one minute.

4.3.3.2 Insulation resistance test. The insulation resistance requirement of 3.4.5.2 shall be tested using a voltmeter having an internal resistance of 30 MΩ and using a 250 volt direct current circuit.

4.3.4 Thermostat test. At an ambient temperature of 68°F to 70°F (20°C to 22°C), the sealing iron or sealing plate, as applicable, shall be heated with its thermostat progressively set at 100°F (56°C) temperature intervals from 200°F to 400°F (93°C to 204°C) and at 450°F (232°) and 500°F (260°C) until each temperature level has been reached, respectively. The temperature at the geometric center of the sealing surface at the moment that the thermostat cuts off at each level shall be checked against its respective thermostat setting to determine compliance with the temperature control requirements of 3.4.3. Any nonconformance shall constitute failure to pass this test.

4.3.5 Handle rigidity test. A force of 5 pounds (22.2 N) shall be applied to the handle of the sealing iron as the sealing surface bears against a flat surface. The stem of the iron shall then be checked to determine stem capability to withstand deflection and deformation. Nonconformance with the deflection and deformation requirements of 3.4.6 shall constitute failure of this test.

4.3.6 Handle temperature test. At an ambient temperature between 68°F and 72°F (20°C and 22°C) and with thermostat set for 450°F (232°C) temperature, the sealing iron shall be heated for one hour with this temperature setting. The temperature of the handle shall then be checked. Nonconformance with the handle temperature requirements of 3.4.6 shall constitute failure of this test.

4.3.7 Wattage test. The wattage of the sealing iron or sealing plate, as applicable, shall be checked with a wattmeter. Nonconformance with the wattage requirements of 3.4.5 or 3.4.7, as applicable shall constitute failure of this test.

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4.3.8 Rest test. The sealing iron rest shall be placed on an 8 X 10 inch (203 X 254 mm) sheet of white bond paper. At an ambient temperature between 68°F and 72°F (20°C and 22°C), and with the thermostat set for 450°F (232°C) temperature, the iron shall be placed on the rest to heat for one hour. The paper shall then be checked for discoloration. Nonconformance with the requirements of 3.4.6.1 shall constitute failure of this test.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each sealing iron or sealing plate shall be packaged in a polyethylene bag conforming to type II, style I of PPP-B-26. Excess air within the bag shall be expelled prior to final closure. Each bagged sealing iron or sealing plate shall then be packaged in a snug-fitting folding or set-up paperboard or fiberboard box conforming to variety 1, style IV, type J, class i of PPP-B-566, or type I, variety 1, class A, style 3 or 4 of PPP-B-676, or style RSC, type CF, (variety SW) or type SF, class domestic grade 200 of PPP-B-636, respectively. Closure shall be in accordance with the appendix of the applicable box specification.

5.1.2 Commercial. Sealing irons or sealing plates shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Thirty type I sealing irons or four type II sealing plates, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in accordance with the appendix of PPP-B-636, alternatively, PPP-B-601.

5.2.2 Level B packing. Unless otherwise specified (see 5.2.2.1), thirty type I sealing irons or four type II sealing plates, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or type SF, class domestic grade 275 of PPP-B-636. Each container shall be closed in accordance with method II of the appendix of PPP-B-636.

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5.2.2.1 Weather-resistant shipping container. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto.

5.2.3 Commercial. Sealing irons or sealing plates, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The sealing iron and sealing plate covered herein are intended for heat-sealing cellophane and polymer film material.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) When electrical characteristics other than specified are required (see 3.4.4).
(d) Whether polytetrafluoroethylene cover for heating surface of sealing iron or sealing plate is required (see 3.4.6 and 3.4.7).
(e) Selection of applicable levels of preservation-packaging and packing (see 5.1 and 5.2).
(f) Whether weather-resistant class fiberboard shipping containers are required for level B shipment (see 5.2.2.1).

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification.

6.4 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

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6.5 Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 99

Preparing activity:

Army - GL

Project No. 3540-0125

Review activity:

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DOCUMENT IDENTIFIER AND TITLE

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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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